A Leakage-Controlled Cold-Start Benchmark for Drug-Combination Synergy Classification on DrugComb v1.5

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Abstract

Background: Drug-combination screening is central to oncology and infectious-disease research, but only a small fraction of possible pairings can be measured experimentally. Most synergy-prediction models rely on rich molecular and omics inputs; it is under-reported how much of the synergy signal is recoverable from pre-treatment identity-and-context descriptors alone. We present a reproducible, leakage-controlled benchmark that asks how well drug-combination synergy *class* can be predicted from compound identities, cell line, tissue, study source, clinical-development phase and nominal target annotations, without using any measured dose-response readout as a feature. Using DrugComb v1.5 (739,964 combination experiments across 26 studies, 17 tissues, 288 cell lines and 4,268 compounds), we define a three-class endpoint from the Zero Interaction Potency (ZIP) score using conventional ± 10 thresholds, and exclude every response-derived variable from the feature set. A multinomial logistic-regression baseline and a class-weighted gradient-boosting (LightGBM) model are evaluated under the predefined grouped split and under leave-drug-out and leave-cell-line-out splits.

Results: On the standard test set, LightGBM attains 0.782 accuracy, 0.805 balanced accuracy, 0.703 macro-F1, 0.545 Cohen's κ and 0.922 macro one-vs-rest AUC, outperforming logistic regression by paired McNemar testing ($p < 0.001$). Precision–recall analysis shows roughly six-fold enrichment of the rare antagonistic and synergistic classes over their priors. However, performance decreases sharply under distribution shift: balanced accuracy falls to 0.694 for unseen drugs and to 0.482 for unseen cell lines, and to the random floor for entirely unseen studies; a no-study ablation shows this is not driven by the study identifier. Permutation importance identifies cell-line identity as the dominant predictor.

Conclusions: Useful synergy triage is achievable for known entities without molecular features, but standard splits partly measure memorization of entity-specific base rates and can materially overstate cold-start deployment performance. The benchmark provides a leakage-controlled lower bound against which richer molecular and omics models can be measured, and argues that cold-start evaluation should be reported alongside conventional random or grouped splits.

Keywords: drug synergy; drug combinations; DrugComb; ZIP score; benchmark; gradient boosting; cold-start generalization; computational pharmacology.

1 Background

Combination therapy—administering two or more drugs simultaneously—is a cornerstone of treatment for cancer, infectious disease, and other complex conditions. When two agents act synergistically the combined effect exceeds what their individual activities would predict, which can improve efficacy at lower doses, suppress the emergence of resistance, and widen the therapeutic window. Antagonistic pairs, conversely, can cancel one another's effects or add unnecessary toxicity. Identifying which of the astronomically many possible pairings are synergistic in a given biological context is therefore a problem of substantial practical value.

The difficulty is combinatorial. With thousands of approved and investigational compounds and hundreds of relevant cell models, the number of drug-pair-by-context experiments far exceeds what any laboratory can screen exhaustively. High-throughput campaigns such as NCI-ALMANAC [1], the O'Neil screen [2], and the AstraZeneca–DREAM challenge [3] have generated hundreds of thousands of measurements, and the DrugComb portal [4, 5] has aggregated and harmonized these into a single resource with standardized synergy scoring. This scale makes data-driven prediction attractive: a model that triages candidate combinations before they reach the bench would let experimental effort concentrate where synergy is most likely.

Most published synergy-prediction models rely on rich molecular inputs—chemical fingerprints or graph representations of the compounds, and transcriptomic, mutational, or copy-number profiles of the cell lines [6, 3]. Such features are powerful but are not always available, are costly to assemble, and complicate deployment. A complementary and under-reported question is *how much of the synergy signal is recoverable from identity and context descriptors alone*: simply knowing which two drugs are paired, in which cell line, tissue, and study, together with each drug's clinical-development phase and catalogued protein targets. Answering this establishes a reproducible lower bound against which structure- and omics-based models should be compared, and yields a lightweight triage model that requires no molecular feature engineering. For readers interested in methods reuse, this benchmark also clarifies what a simple identity-and-context model can and cannot achieve before more expensive molecular descriptors are added.

We address that question at scale using the DrugComb v1.5 modeling release. The contribution is not a new machine-learning algorithm; rather, it is a reusable benchmark and cautionary evaluation study. We provide: (i) a leakage-controlled context-only synergy-classification task on a large public dataset, with every response-derived column excluded from the inputs; (ii) paired evaluation of a transparent linear baseline and a non-linear gradient-boosted model using bootstrap confidence intervals, precision–recall analysis suited to the class imbalance, calibration diagnostics and cardinality-robust permutation importance; and (iii) explicit leave-drug-out and leave-cell-line-out experiments that quantify how far identity-and-context signals transfer to unseen compounds and unseen cellular backgrounds. The last point is central: aggregate performance on a conventional split can materially overstate performance in cold-start deployment.

1.1 Related work

Synergy quantification rests on reference null models for non-interaction. The Loewe additivity model [7] assumes a drug cannot interact with itself; the Bliss independence model [8] assumes probabilistic independence of effects; the Highest Single Agent (HSA) model uses the stronger single agent as reference; and the Zero Interaction Potency (ZIP) score [9] combines the Loewe and Bliss assumptions and is comparatively robust to noise, which motivates its use as our target. On the modeling side, DeepSynergy [6] demonstrated that neural networks on chemical and genomic features predict O'Neil synergy well, and the AstraZeneca–DREAM community effort [3] benchmarked many such approaches. Recent work has continued toward richer representations, including conventional machine-learning pipelines for FDA-approved cancer drugs [10], multi-view drug and cell-line representations with multi-task attention [11], graph-neural-network approaches summarized in recent reviews [12], and image-based molecular and transcriptomic encodings such as SynergyImage [13]. These methods depend on molecular, graph, image or omics descriptors;

the present work deliberately removes them to isolate and quantify the identity-and-context signal. This design makes the study a lower-bound benchmark rather than a replacement for molecular models. It also allows a direct test of whether conventional splits reward knowledge of frequently assayed entities more than transferable pharmacological signal.

2 Methods

2.1 Dataset

All experiments use the DrugComb v1.5 modeling dataset, a cleaned and harmonized derivative of the DrugComb portal summary release. The synergy-prediction table contains 739,964 drug-pair combination experiments. Each row records a pair of compounds (designated drug A and drug B) tested against a named cell line, annotated with the originating study and tissue of origin. The data span 26 studies, 17 tissues, 288 cell lines, and 4,268 unique compounds. The largest studies are NCI-ALMANAC (311,604 rows), Friedman (198,722), and O'Neil (92,208); the most represented tissues are skin (287,952), haematopoietic/lymphoid (78,801), and lung (68,345), reflecting the melanoma- and leukaemia-heavy composition of the source screens.

2.2 Outcome definition and labels

The target is the synergy class derived from the ZIP synergy score, which quantifies the excess combination effect relative to the expectation under a reference null model. Following the dataset convention and the wider literature, the continuous ZIP score is discretized into three classes: a pair is *synergistic* when ZIP $\geq +10$, *antagonistic* when ZIP ≤ -10, and *additive* otherwise. Across the full dataset this yields a strongly imbalanced distribution of 567,236 additive (76.7%), 93,379 synergistic (12.6%), and 79,349 antagonistic (10.7%) combinations (Figure 1). This imbalance is itself a defining feature of the synergy-discovery problem and motivates the class-balanced training and macro-averaged, prior-aware evaluation described below.

2.3 Features and leakage control

The guiding principle is that the model must predict from information available *before* the combination is assayed. We use five categorical descriptors—the two drug identities, the cell line, the tissue, and the study—together with each compound's clinical-development phase (ordinal, 0 pre-clinical to 4 approved) and descriptors derived from each drug's catalogued protein targets: the number of annotated targets per compound, indicator flags for whether any target is recorded, and a binary flag for whether the two drugs share at least one target. Missing clinical-phase values are encoded with a distinct sentinel; missing target annotations contribute a target count of zero.

Crucially, every column that constitutes a measured response of the same screen was withheld from the feature set to avoid target leakage. This includes the three alternative synergy scores (Loewe, HSA, Bliss) and their derived labels, the combination sensitivity score (CSS), relative inhibition (RI), single-agent and combination half-maximal inhibitory concentration (IC50) values, and the S-score family. Because these are computed from the same dose-response surface as the ZIP target, including any of them would let the model reconstruct the label rather than learn a predictive relationship.

2.4 Encoding

Two encodings were used, matched to the two model families. For logistic regression, each categorical variable is represented by a frequency encoding plus a smoothed mean-target encoding for each class, fitted on the training partition only (additive smoothing with pseudo-count 20 toward the global prior; unseen levels fall back to the prior); all features are then standardized. For LightGBM, the categorical variables are passed to its native categorical handling, which learns partitions of category levels directly and avoids the dimensional explosion of one-hot encoding for the high-cardinality drug and cell-line fields.

2.5 Evaluation regimes and data splitting

We evaluate under three regimes designed to probe different forms of generalization (Table 1). **(1) Standard:** the dataset's predefined train/validation/test split, which assigns combinations deterministically by study, cell line, and drug pair so that near-duplicate measurements of the same pair-context do not straddle partitions; the resulting partitions hold 592,068 / 74,278 / 73,618 combinations. Note that drugs, cell lines, and studies still appear on both sides of this split—the model is tested on novel *pairings* of known entities. **(2) Leave-drug-out (cold-drug):** 12% of unique compounds are held out; the test set is every combination involving at least one held-out drug, and the model trains only on combinations where both drugs are seen. **(3) Leave-cell-line-out (cold-cell):** 15% of cell lines are held out and form the test set, with training restricted to the remaining cell lines. Regimes (2) and (3) emulate the realistic deployment cases of scoring a *new* compound or a *new* cellular background. We note that the cold-cell regime holds out cell lines but not tissues, so a held-out cell line may belong to a tissue represented in training; the cold-cell result should therefore be read as an upper bound on true cell-line transfer.

To address study and tissue confounding, we additionally performed three robustness checks after the primary analysis: a no-study ablation on the standard split, a leave-study-out split, and a leave-tissue-out split. For leave-study-out and leave-tissue-out evaluation, whole categories were sampled in a fixed random order (seed 42) until approximately 15% of rows or more were assigned to the held-out test set; the remaining rows were split into training and validation sets with a 90/10 training/validation split. Because large screens are highly imbalanced, this rule held out 14 of 26 studies (including NCI-ALMANAC) and 7 of 17 tissues.

Table 1: Evaluation regimes. Standard is the dataset's grouped split; cold-drug and cold-cell hold out entities entirely.

Regime	Train	Validation	Test	Held-out entity
Standard (grouped)	592,068	74,278	73,618	none (novel pairings)
Leave-drug-out	540,423	60,046	139,495	512 drugs (12%)
Leave-cell-line-out	559,653	62,183	118,128	43 cell lines (15%)

2.6 Models

Two classifiers are trained. The first is a multinomial logistic regression (L2-regularized, inverse regularization strength $C = 1.0$) with balanced class weights, a transparent linear baseline. The second is a LightGBM [14] gradient-boosted decision-tree ensemble for multiclass classification (up to 350 trees, learning rate 0.1, 64 leaves, minimum 100 samples per leaf, feature and row subsampling 0.8, L2 penalty 2.0, maximum 255 histogram bins), with validation-set monitoring of multiclass log-loss and early stopping (patience 30 rounds). To counter the imbalance, both models use class weights inversely proportional to class frequency—for LightGBM applied as per-sample weights—so the rare classes contribute comparably to the abundant additive class. All hyperparameters were fixed a priori or selected using the training and validation partitions only; the test partition was never used during model development.

2.7 Metrics

All metrics are computed on held-out data untouched during development. Because the classes are imbalanced and the minority classes are of greatest interest, we emphasize class-balanced and prior-aware measures: balanced accuracy (mean per-class recall), macro-F1, Cohen's κ, macro one-vs-rest (OvR) receiver-operating-characteristic AUC, and—most appropriate under imbalance—area under the precision–recall curve (AUPRC) for each class relative to its prior. We report 95% bootstrap confidence intervals (300 resamples of the test set) and compare the two models with a McNemar test on paired correctness. We additionally assess probability calibration via reliability diagrams and assign feature importance by permutation (the decrease in macro-F1 when a feature is shuffled), which, unlike split-gain importance, is not biased toward high-cardinality variables.

2.8 Implementation

Data processing used pandas; models used scikit-learn [15] and LightGBM 4.6. A fixed seed (42) was used throughout. The full pipeline—feature construction, training, the three evaluation regimes, bootstrap and permutation analyses, and figure generation—is organized as reproducible scripts so that all reported numbers can be regenerated from the public dataset (see Availability of data and materials).

3 Results

3.1 Dataset characteristics

The class distribution (Figure 1) confirms the strong dominance of additive combinations (76.7%), with synergistic and antagonistic pairs at 12.6% and 10.7%. A trivial model predicting "additive" for every pair would reach 76.6% accuracy on the test set while being useless for finding synergy; this is why class-balanced and prior-aware metrics anchor the evaluation.

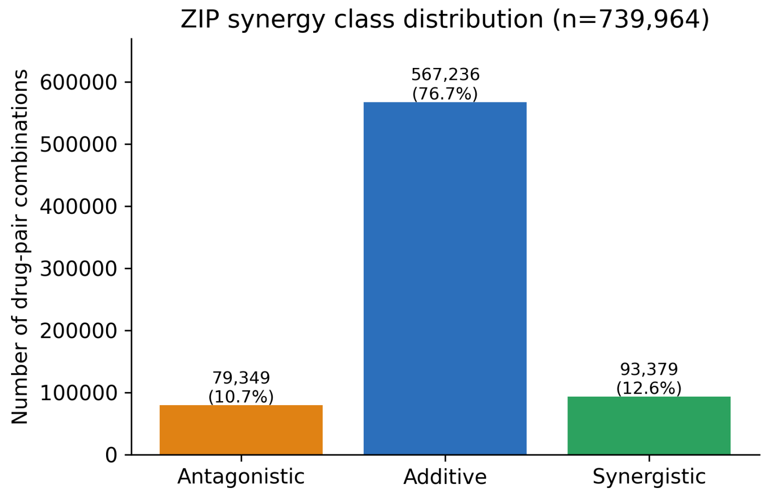

Figure 1: Distribution of ZIP synergy classes across all 739,964 combinations. Additive pairs dominate; synergistic and antagonistic combinations each form roughly one-eighth of the data.

3.2 Overall performance on the standard split

Table 2 summarizes test performance with bootstrap confidence intervals. LightGBM outperforms the logistic baseline on every metric, and by a wide margin on the class-balanced ones. Although its raw accuracy (0.782) is only modestly above the 0.766 majority baseline, its balanced accuracy (0.805) and macro-F1 (0.703) far exceed the baseline's 0.333 and 0.289, showing the headline accuracy comes from recovering minority classes rather than defaulting to the majority. The confidence intervals for the two models do not overlap on any metric, and a McNemar test is decisive: LightGBM is uniquely correct on 9,941 test cases versus 3,130 for logistic regression ($\chi^2 = 3548$, $p < 0.001$).

Table 2: Standard-split test performance (73,618 combinations) with 95% bootstrap confidence intervals. Best value per row in bold.

Metric	Majority baseline	Logistic Regression	LightGBM
Accuracy	0.766	0.689 [0.686, 0.692]	**0.782** [0.779, 0.785]
Balanced accuracy	0.333	0.633 [0.629, 0.639]	**0.805** [0.801, 0.809]
Macro-F1	0.289	0.559 [0.555, 0.564]	**0.703** [0.699, 0.706]
Cohen's κ	0.000	0.373 [0.367, 0.379]	**0.545** [0.540, 0.551]
Macro AUC (OvR)	—	0.821 [0.818, 0.824]	**0.922** [0.920, 0.924]

3.3 Per-class behavior

The per-class breakdown (Table 3) and confusion matrices (Figure 2) show where the models succeed. With class re-weighting, LightGBM recovers the rare classes at high recall—84.1% of antagonistic and 80.5% of synergistic pairs—while keeping 77.0% recall and 94.9% precision on additive pairs. The cost is moderate precision on the rare classes (0.52 antagonistic, 0.49 synergistic): some additive pairs are flagged as interacting. Most residual errors are confusions between an interacting class and additive, rather than between the two opposite interaction types—an encouraging pattern for triage.

Table 3: Per-class precision, recall and F1 on the standard test split.

Class	Model	Precision	Recall	F1	Support
Antagonistic	LightGBM	0.524	0.841	0.646	7,989
	Logistic	0.354	0.574	0.438	7,989
Additive	LightGBM	0.949	0.770	0.850	56,395
	Logistic	0.923	0.719	0.809	56,395
Synergistic	LightGBM	0.493	0.805	0.612	9,234
	Logistic	0.334	0.607	0.431	9,234

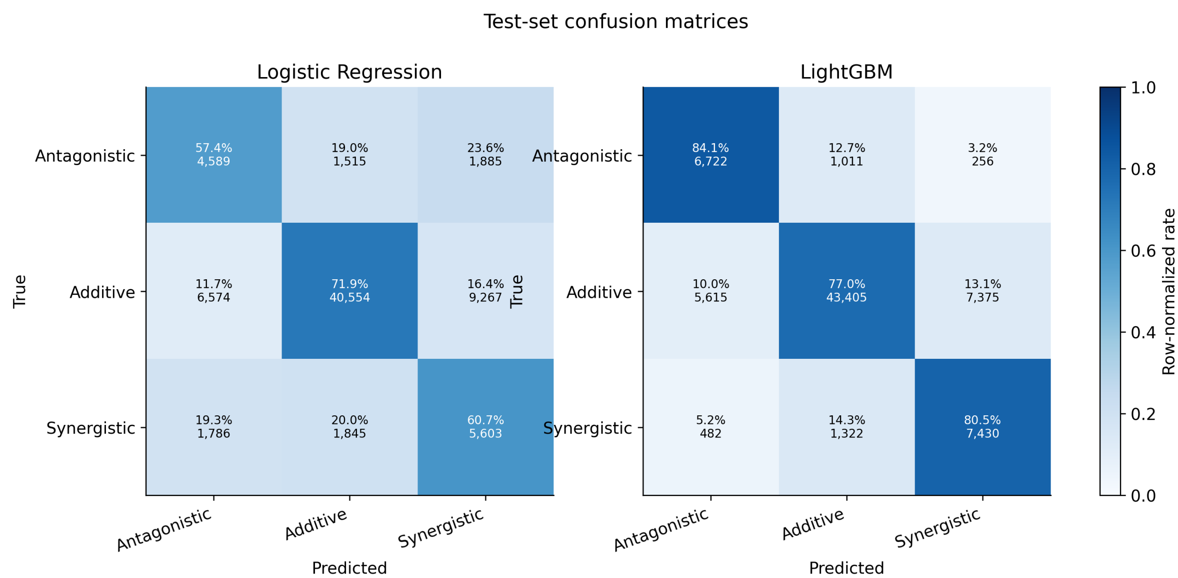

Figure 2: Row-normalized confusion matrices. LightGBM (right) places far more mass on the diagonal for the rare classes than the logistic baseline (left).

3.4 Precision–recall and calibration

Because ROC-AUC is optimistic under strong imbalance, we report precision–recall analysis (Figure 3). LightGBM achieves AUPRC of 0.71 (antagonistic) and 0.72 (synergistic), against class priors of 0.11 and 0.13—roughly a six-fold enrichment—and 0.96 for additive. The logistic baseline trails substantially on the minority classes (0.48 and 0.37). Reliability diagrams (Figure 4) show the predicted probabilities are reasonably calibrated for the additive class but, as expected after class re-weighting, over-estimate the probability of the rare classes; a post-hoc recalibration would be advisable before using the scores as absolute risks.

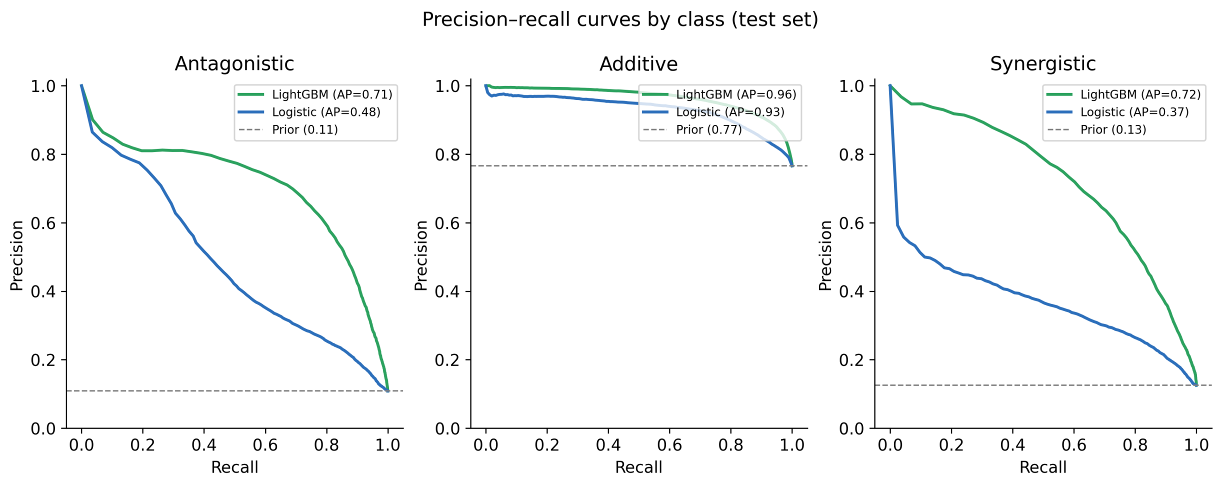

Figure 3: Per-class precision–recall curves on the standard test split, with average precision (AP) and the class prior (dashed). LightGBM enriches the rare classes roughly six-fold over their priors.

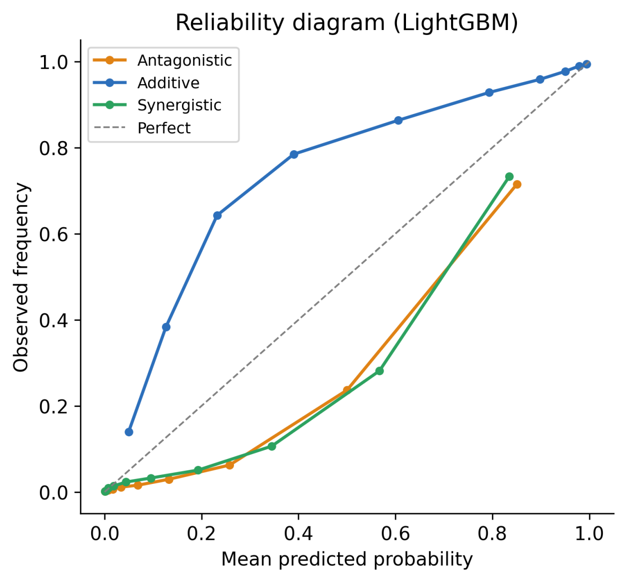

Figure 4: Reliability diagram for LightGBM. The additive class is well calibrated; the re-weighted rare classes are over-confident and would benefit from post-hoc recalibration.

3.5 Feature importance

Permutation importance (Figure 5), which is robust to feature cardinality, identifies the *cell line* as the single most informative feature (macro-F1 drop 0.247 when shuffled), ahead of the two drug identities (0.128 and 0.115) and the study (0.017). Notably, this re-orders the naive gain-based ranking, in which the high-cardinality drug fields appeared dominant—illustrating why cardinality-robust attribution matters. The engineered target and clinical-phase descriptors contribute little in this feature set, and tissue is marginally redundant given the cell line. Study identity contributes modestly; because it can encode assay platform and curation effects, it should be treated as a potential confounder rather than a purely biological predictor. The primacy of cell line foreshadows the cold-start result below: the model relies heavily on having seen the cellular background during training.

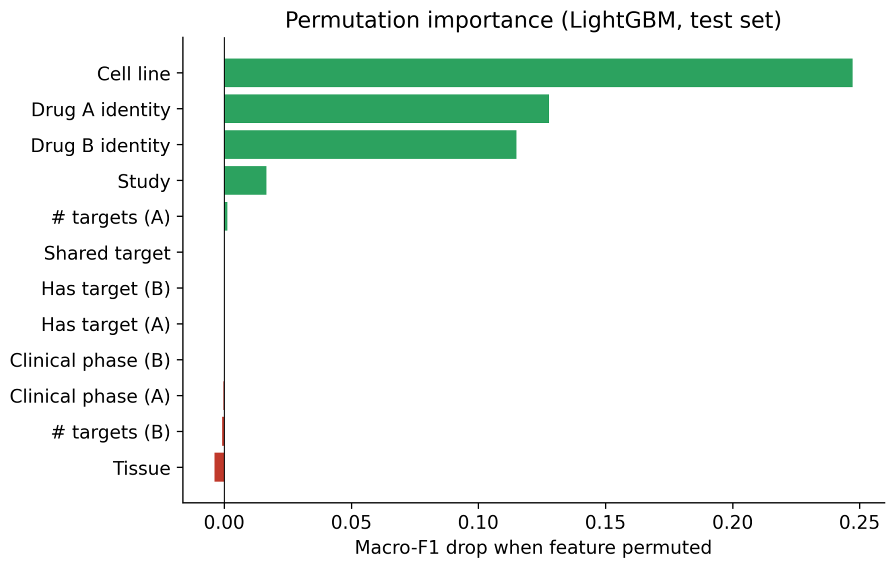

Figure 5: Permutation importance (decrease in macro-F1 when each feature is shuffled). Cell line dominates, followed by the two drug identities; coarse target and clinical-phase descriptors contribute little in this feature set.

3.6 Cold-start generalization

The central new result concerns distribution shift (Figure 6, Table 4). When the test set involves drugs unseen in training, LightGBM's balanced accuracy falls from 0.805 to 0.694 and macro-F1 from 0.703 to 0.631—a clear but graceful degradation: the model still recovers most antagonistic (70%) and a majority of synergistic (59%) pairs, indicating some genuine transfer through cell-context and target features. When the test set involves *unseen cell lines*, however, performance collapses: balanced accuracy drops to 0.482, barely above the 0.333 random floor, with antagonistic recall falling to 0.20. In other words, the strong standard-split numbers are substantially carried by memorization of per-cell-line synergy base rates, and the model has limited ability to generalize to a cellular background it has never seen. This is the gap that molecular and omics features would need to close, and it is invisible to the conventional split.

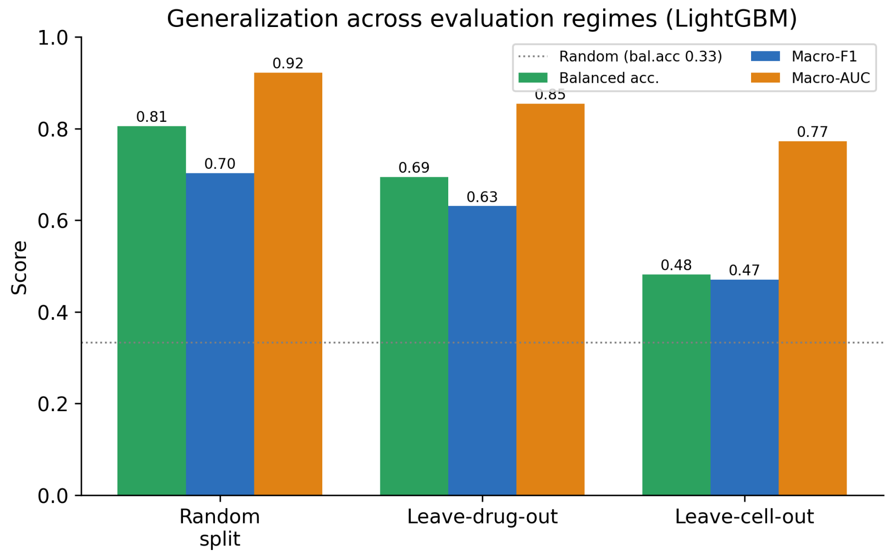

Figure 6: LightGBM generalization across the three regimes. Performance degrades for unseen drugs and collapses toward the random floor for unseen cell lines.

Table 4: LightGBM performance across evaluation regimes, with per-class recall.

				Recall		
Regime	Bal. acc.	Macro-F1	Macro-AUC	Antag.	Add.	Syn.
Standard (grouped)	**0.805**	**0.703**	**0.922**	0.841	0.770	0.805
Leave-drug-out	0.694	0.631	0.855	0.696	0.792	0.594
Leave-cell-line-out	0.482	0.470	0.773	0.196	0.804	0.445

3.7 Study confounder and cross-study/tissue transfer

Three further experiments probe whether the standard-split performance depends on the study confounder and how far it transfers across whole studies and tissues (Table 5). First, a no-study ablation—retraining with the study feature removed—leaves standard-split performance essentially unchanged (balanced accuracy 0.807 $\rightarrow$ 0.806, macro-F1 unchanged at 0.708), so the reported numbers are not an artefact of study-level platform or curation signal. Second, under leave-study-out evaluation, where 14 of the 26 studies (including NCI-ALMANAC) are held out entirely, performance collapses to the random floor (balanced accuracy 0.332, macro one-vs-rest AUC 0.421), with the model defaulting almost entirely to the additive class (synergistic recall 0.01). Third, leave-tissue-out evaluation (7 of 17 tissues held out) gives balanced accuracy 0.431 and AUC 0.772—a collapse comparable to the leave-cell-line-out regime. Together these results show essentially no transfer to unseen studies and limited transfer to unseen tissues, confirming that the strong standard-split numbers reflect entity- and context-specific base rates rather than mechanism, while ruling out the study identifier itself as the driver.

Table 5: Robustness experiments (LightGBM, fixed seed 42): no-study ablation on the standard split, and transfer to entirely unseen studies and tissues. Held-out counts are given as (held/total).

				Recall		
Regime	Bal. acc.	Macro-F1	Macro-AUC	Antag.	Add.	Syn.
Standard, with study	0.807	0.708	0.923	0.840	0.778	0.802
Standard, no study	0.806	0.708	0.922	0.838	0.778	0.803
Leave-study-out (14/26)	0.332	0.312	0.421	0.000	0.985	0.010
Leave-tissue-out (7/17)	0.431	0.466	0.772	0.149	0.976	0.169

3.8 Per-tissue performance

Disaggregating the standard-split results by tissue (Figure 7) shows that aggregate metrics are not uniform: balanced accuracy varies markedly across tissues, and the global figure is dominated by the few large, well-represented tissues. Tissues with abundant training data tend to be predicted more reliably, consistent with the entity-memorization picture.

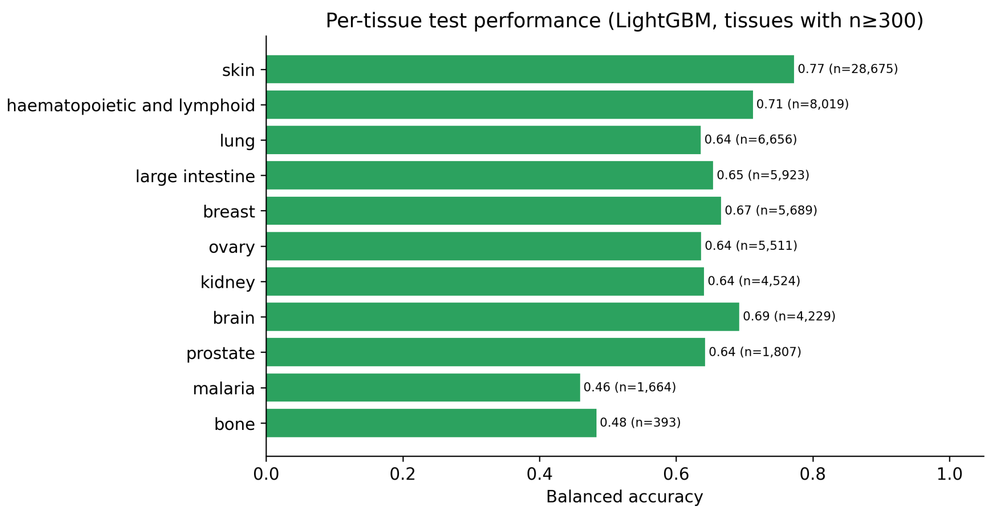

Figure 7: Per-tissue balanced accuracy on the standard test split (tissues with $n \geq 300$ test rows). Performance is heterogeneous across tissues.

4 Discussion

Three findings stand out. First, drug-combination synergy class is predictable above chance from pre-treatment descriptors alone: without chemical structures or omics profiles, a gradient-boosted model reaches 0.805 balanced accuracy and enriches the rare interacting classes over their priors on a large, leakage-controlled benchmark. Second, the comparison between models is informative: the logistic baseline clears the majority reference on balanced metrics, but the additional gains from LightGBM (+0.17 balanced accuracy) indicate non-linear drug-by-cell-context interactions that a linear model cannot capture. Third, and most importantly, the cold-start analysis shows that this performance does not transfer uniformly. The model is useful for novel pairings of known drugs in known cell lines, partially robust to unseen drugs, much less reliable for unseen cell lines or unseen tissues, and no better than chance for entirely unseen studies (Table 5).

This reframes the result for computational-biology and bioinformatics readers. The model should not be interpreted as a mechanistic discovery system or as a replacement for molecular-feature models. It is a lightweight baseline and diagnostic tool: if a more complex chemical or multi-omics model cannot improve on this benchmark, especially under leave-drug-out or leave-cell-line-out evaluation, its added complexity is difficult to justify. Conversely, the cold-cell failure identifies the regime where molecular and cellular features are most needed. We emphasize that the claim that molecular and omics features would close this gap is a motivated hypothesis rather than a demonstrated result, because no molecular model is run here; testing it directly is the natural next step.

4.1 Limitations

Several caveats temper interpretation. (1) Features encode identity, not mechanism; the cold-cell collapse quantifies the resulting ceiling. (2) Study identity may encode assay platform, dose design, laboratory and curation effects, and should not be interpreted as a biological variable. A no-study ablation leaves standard-split performance essentially unchanged, but leave-study-out evaluation collapses to the random floor (Table 5), so cross-study deployment cannot be assumed. (3) The cold-cell collapse—part of any residual signal and part of the collapse—is largely a context effect: leave-tissue-out evaluation, holding out whole tissues, degrades comparably (balanced accuracy 0.431; Table 5), indicating limited transfer to unseen tissues as well as unseen cell lines. (4) The dataset is composition-skewed toward a few large screens and toward skin and blood-cancer tissues, so aggregate metrics are dominated by those settings. (5) The ZIP score and ± 10 thresholds, though standard, are one of several synergy definitions, and pairs near a threshold are intrinsically ambiguous; some residual error reflects label noise. (6) After class re-weighting the predicted probabilities are not calibrated for the rare classes (Figure 4). (7) ZIP values in the source data include extreme outliers, suggesting residual assay-quality variation that classification only partly absorbs. (8) Drug A/drug B ordering is treated as informative although synergy is symmetric in the pair; an order-invariant encoding or swapped-pair augmentation should be evaluated.

4.2 Future work

The natural extensions follow directly from the limitations. The no-study, leave-study-out and leave-tissue-out analyses reported above (Table 5) already address the study confounder and cross-study/tissue transfer; the immediate remaining robustness check is an order-invariant drug-pair encoding. A natural next comparison is to reuse these exact splits with Morgan-fingerprint, cell-line omics, molecular-image or learned graph-embedding models to quantify how much chemical structure and cellular biology add above the identity-only ceiling, especially in the cold-cell and leave-study-out regimes where the present model fails. Recent multimodal, graph-based and image-based models provide natural comparators for this next stage [11, 12, 13]. Calibrating predicted probabilities (e.g. temperature scaling on a held-out fold) would make the scores usable as absolute risks, and treating the task as ordinal regression or continuous ZIP-score prediction would better match a ranked-triage workflow. The benchmark and pipeline released here are intended as the reference point for these comparisons.

5 Conclusions

We presented a large-scale, leakage-controlled benchmark for classifying drug-combination synergy from compound and cell-context descriptors on DrugComb v1.5 (739,964 combinations). A class-weighted LightGBM model reached 0.782 accuracy, 0.805 balanced accuracy, 0.703 macro-F1 and 0.922 macro AUC on the standard split, outperforming a logistic baseline ($p < 0.001$) and enriching the rare interacting classes over their priors. Permutation importance identified the cell line as the dominant predictor. Crucially, cold-start evaluation revealed that balanced accuracy falls to 0.69 for unseen drugs, to 0.48 for unseen cell lines, to 0.43 for unseen tissues and to the random floor for entirely unseen studies, exposing reliance on entity-specific information that conventional splits can hide; a no-study ablation confirms this is not an artefact of the study identifier. The work demonstrates that useful synergy triage is achievable for known entities without molecular features. It also supplies a reproducible baseline against which richer models can be measured, and argues that cold-start evaluation should be standard practice in this field.

Abbreviations

AUC: area under the (receiver-operating-characteristic) curve; AUPRC: area under the precision–recall curve; CSS: combination sensitivity score; DREAM: Dialogue for Reverse Engineering Assessments and Methods; HSA: highest single agent; IC50: half-maximal inhibitory concentration; NCI-ALMANAC: National Cancer Institute A Large Matrix of Anti-Neoplastic Agent Combinations; OvR: one-vs-rest; RI: relative inhibition; ROC: receiver operating characteristic; ZIP: zero interaction potency.

Ethics approval and consent to participate

Not applicable. This study used only publicly available, de-identified in vitro drug-combination screening data and did not involve new studies of human participants, human tissue or animals.

Consent for publication

Not applicable.

Availability of data and materials

The source DrugComb v1.5 summary file is publicly available from the DrugComb download portal (https://drugcomb.org/download/) and from Zenodo record 15235991 (https://zenodo.org/records/15235991). The modeling table used in this study was generated locally from the public DrugComb summary file using the preprocessing script described in the Methods; no restricted or patient-identifiable data were used. The analysis code (preprocessing, training, the three evaluation regimes, the no-study, leave-study-out and leave-tissue-out robustness experiments, a Morgan-fingerprint structure-baseline comparator, and the bootstrap, permutation and figure-generation scripts) and the metric outputs needed to reproduce all reported numbers are included as supplementary material with this submission package and are intended for deposition in a public repository before final journal submission. Programming language: Python ($\geq$ 3.10). All experiments were run with a fixed random seed (42).

Competing interests

The author declares that they have no competing interests.

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Authors' contributions

X.P. designed the study, prepared the dataset, implemented the models, performed the analysis and wrote the manuscript. The author read and approved the final manuscript.

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